

# Ho Fung Tsoi

Postdoctoral Researcher

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## Research Interests

Building machine learning methods for accelerating scientific discoveries at particle physics experiments.  
**Experimental particle physics:** new physics searches (SUSY, exotic Higgs) and trigger system at the CERN LHC  
**Machine learning:** self-supervised pre-training methods, anomaly detection, symbolic regression  
**Fast inference on FPGAs:** low-latency (sub-microsecond) machine learning algorithms on FPGAs

## Education

**University of Wisconsin–Madison**, Ph.D. Physics 2024  
Thesis: [Search for exotic Higgs boson decays with CMS and fast machine learning solutions for the LHC](#)  
Advisor: Sridhara Dasu

**The Chinese University of Hong Kong**, B.Sc. Physics 2018  
Thesis: First passage time problem of the time-dependent Ornstein-Uhlenbeck process  
Advisor: Chi-Fai Lo  
Visiting student at University of California Berkeley & Berkeley Lab (Jan – Aug 2017)

## Positions

**Postdoctoral Researcher**, University of Pennsylvania (Supervisor: Dylan Rankin) 2024 – Present

## Awards

CERN, [CMS PhD Thesis Award](#) 2024  
UWisconsin, Cornelius P. and Cynthia C. Browne Fellowship 2024  
CUHK, CN Yang Scholarship 2017  
CUHK, Wei Lun Foundation Exchange Scholarship 2017  
CUHK, Professor Dennis Yam Kuen Lo Physics Award 2016

## Selected Research Experience

### Search for new physics at the LHC

- **SUSY.** ATLAS Run 3 search for electroweak production of supersymmetric particles in compressed mass spectra with low-momentum tracks (2026–Present).
- **Exotic Higgs.** CMS Run 2 search for exotic Higgs boson decays into pseudoscalars in the final state of two b quarks and two  $\tau$  leptons, and the combination with the  $\mu\mu b\bar{b}$  final state [4] (2020–2024).

### Novel ML methods for high energy physics

- **Self-supervised learning.** Self-distillation pre-training method **jBOT** for jets [14]; VICReg pre-training to improve resonance mass width reconstruction; data-driven formation of invariant training pairs using energy mover's distance (2025–Present).
- **SymbolFit.** Automatic discovery of parametric functions to model HEP data [6,13] (2024–2026).

### Low-latency (sub-microsecond) ML algorithms on FPGAs

- **hls4ml.** Co-developer; an open-source tool that converts and optimizes trained neural networks into FPGA firmware [7] (2023–Present).
- **SpikePID.** Neuromorphic spiking neural networks (SNNs) for on-detector  $dN/dx$  clustering counting in future drift chambers [11] (2025–Present).
- **SparsePixels.** CNNs with sparse convolution for spatially sparse image data on FPGAs [8] (2024–Present).
- **SymbolNet.** Compression of neural networks using symbolic regression for fast inference [3,5] (2022–2024).

### Trigger system at the LHC

- **GNTau.** Commissioning the GNN-based tau trigger at the ATLAS High-Level Trigger for Run 3 [9] (2024–2025).
- **CICADA.** Development of an anomaly detection trigger algorithm at the CMS Level-1 Trigger [10] (2022–2024).
- Development of the Data Quality Monitoring (DQM) software for the Calorimeter Layer-1 subsystem at the CMS Level-1 Trigger (2020–2024).
- HL-LHC trigger sensitivity projection for the CMS searches for exotic Higgs boson decays [1] (2020–2021).

## Selected Publications

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Listed below are selected publications to which I was a primary author. I also co-authored papers by [CMS](#) and [ATLAS](#) as a collaboration member. Complete list at [INSPIRE](#). Preprints at [arXiv](#).

### 2026

14. “jBOT: Semantic Jet Representation Clustering Emerges from Self-Distillation”  
[Ho Fung Tsoi](#), Dylan Rankin [SciPost Phys.](#)  
[arXiv:2601.11719](#)
13. “Machine Can Automatically Discover Parametric Functions to Model HEP Data”  
[Ho Fung Tsoi](#), Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles [ICHEP 2026](#)  
[arXiv:2607.19750](#)
12. “SparsePixels+ +: Scalable Sparse Convolution on FPGAs”  
[Ho Fung Tsoi](#), Dylan Rankin, Philip Harris *Preparing for sub.*
11. “SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers”  
[Ho Fung Tsoi](#), Kam Wai Lai, Dylan Rankin *Preparing for sub.*
10. “Unsupervised Anomaly Detection for Real-Time Data Acquisition in the CMS Experiment at the LHC”  
CMS Collaboration *Preparing for sub.*  
[CMS PAS](#)
9. “Performance of the ATLAS Tau Trigger in Run 3”  
ATLAS Collaboration *Preparing for sub.*

### 2025

8. “SparsePixels: Efficient Convolution for Sparse Data on FPGAs”  
[Ho Fung Tsoi](#), Dylan Rankin, Vladimir Loncar, Philip Harris [Mach. Learn. Sci. Tech.](#)  
[arXiv:2512.06208](#)
7. “hls4ml: A Flexible, Open-Source Platform for Deep Learning Acceleration on Reconfigurable Hardware”  
Jan-Frederik Schulte et al. [ACM TRETS](#)  
[arXiv:2512.01463](#)

### 2024

6. “SymbolFit: Automatic Parametric Modeling with Symbolic Regression”  
[Ho Fung Tsoi](#), Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles, Vladimir Loncar [Comput. Softw. Big Sci.](#)  
[arXiv:2411.09851](#)
5. “SymbolNet: Neural Symbolic Regression with Adaptive Dynamic Pruning for Compression”  
[Ho Fung Tsoi](#), Vladimir Loncar, Sridhara Dasu, Philip Harris [Mach. Learn. Sci. Tech.](#)  
[arXiv:2401.09949](#)
4. “Search for Exotic Decays of the Higgs Boson to a Pair of Pseudoscalars in the  $\mu\mu b\bar{b}$  and  $\tau\tau b\bar{b}$  Final States”  
CMS Collaboration [Eur. Phys. J. C](#)  
[arXiv:2402.13358](#)

### 2023

3. “Symbolic Regression on FPGAs for Fast Machine Learning Inference”  
[Ho Fung Tsoi](#), Adrian Alan Pol, Vladimir Loncar, Ekaterina Govorkova, Miles Cranmer, Sridhara Dasu, Peter Elmer, Philip Harris, Isobel Ojalvo, Maurizio Pierini [CHEP 2023](#)  
[arXiv:2305.04099](#)
2. “Searches for exotic Higgs boson decays with the CMS experiment”  
[Ho Fung Tsoi](#) (on behalf of the CMS Collaboration) [EPS-HEP 2023](#)

### 2021

1. “The Phase-2 Upgrade of the CMS Data Acquisition and High Level Trigger”  
CMS Collaboration [CMS Technical Design Report](#)

## Conference, Workshop, Seminar Presentations

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| Fast Machine Learning for Science Conference (FastML) — UCSD, USA<br><a href="#">SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers</a><br><a href="#">SparsePixels+ +: Scalable Sparse Convolution on FPGAs</a> | Sep 2026 |
| International Conference on High Energy Physics (ICHEP) — Natal, Brazil<br><a href="#">Machine Can Automatically Discover Parametric Functions to Model HEP Data</a>                                                           | Jul 2026 |

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| TREASURE: Tokenizing HEP Collider Data for AI Discovery — BNL, USA<br><a href="#">Self-Supervised Learning: Machine Can Learn a Lot by Just Observing</a>                                         | Apr 2026   |
| Coordinating Panel on Advanced Detectors (CPAD) — UPenn, USA<br><a href="#">SparsePixels: Efficient Convolution for Sparse Data on FPGAs</a>                                                      | Oct 2025   |
| 2024 CMS Thesis Award Ceremony Talk (CMS Week) — CERN, Switzerland<br><a href="#">Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC</a>               | Oct 2025   |
| Fast Machine Learning for Science Conference (FastML) — ETH Zurich, Switzerland<br><a href="#">SparsePixels: Efficient Convolution for Sparse Data on FPGAs</a>                                   | Sep 2025   |
| High-energy physics seminar — UPenn; U.Washington; Fermilab, USA<br><a href="#">Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC</a>                 | 2023, 2024 |
| US LHC Users Association Meeting (US LUA) — Fermilab, USA<br><a href="#">CICADA: Anomaly Detection for New Physics Searches at the CMS Level-1 Trigger</a>                                        | Dec 2023   |
| Machine Learning at Level-1 Trigger Workshop (ML@L1) — CERN, Switzerland<br>Anomaly Detection – CICADA: Status, Plans, and Prospects for Phase-2                                                  | Dec 2023   |
| CMS Machine Learning Town Hall — CERN, Switzerland<br>L1 Anomaly Detection with Calorimeter Inputs: Status and Opportunities                                                                      | Sep 2023   |
| European Physical Society Conference on High Energy Physics (EPS-HEP) — Hamburg, Germany<br><a href="#">Searches for exotic Higgs boson decays with the CMS experiment</a>                        | Aug 2023   |
| International Conference on Computing in High Energy & Nuclear Physics (CHEP) — Norfolk VA, USA<br><a href="#">Symbolic Regression on FPGAs for Fast Machine Learning Inference</a>               | May 2023   |
| International Conference on Applied Mathematics — CityU, Hong Kong<br>First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process: a Model for Stochastic Decision-Making Process | May 2016   |

## Professional Services

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### Membership

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| • Member, <a href="#">ATLAS Collaboration</a> , CERN                                                      | 2024 – Present |
| • Affiliated member, <a href="#">Accelerated AI Algorithms for Data-Driven Discovery (A3D3) Institute</a> | 2024 – Present |
| • Member, <a href="#">CMS Collaboration</a> , CERN                                                        | 2020 – 2024    |

### Collaboration Role

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|--------------------------------------------------------------------------------------------|-------------|
| • <b>Co-convenor</b> , SUSY Monte Carlo & Interpretation subgroup [CMS]                    | 2023 – 2024 |
| • <b>Contact person</b> , Higgs boson Monte Carlo group [CMS]                              | 2021 – 2023 |
| • <b>Software developer</b> , Data Quality Monitoring system for CaloLayer-1 trigger [CMS] | 2020 – 2024 |

### Journal Referee

*Transactions on Machine Learning Research* (2026); *IEEE Transactions on Evolutionary Computation* (2026); *Wiley Advanced Theory and Simulations* (2025).

## Teaching and Mentoring

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### Teaching

- Guest lecturer [UPenn], Data Analysis for the Natural Sciences II: Machine Learning (PHYS 3359, Spring 2025)
- Teaching assistant [U.Wisconsin], undergraduate physics courses (Fall 2018 – Spring 2020)

### Master students

- Kam Wai Lai [UPenn]: Spiking neural network on FPGAs for cluster counting in drift chambers (2025–2026)
- Yuyan Wang [UPenn]: Joint-Embedding Predictive Architecture (JEPAs) for jet anomaly detection (2025–2026)

### Undergraduate students

- Chenjia Ni [UPenn]: Lorentz-equivariant architecture on FPGAs (2026–Present)
- Abhay Agarwal [UPenn]: Scalable sparse convolution on FPGAs (2026–Present)
- Alex Yang [UPenn]: Mass regression of supersymmetric particles with contrastive learning (2025)
- Ashni Kumar [Drexel]: Acceleration of reinforcement learning agent inference on FPGAs for astronomical observations (2025)